Mathematical and Computational Methods

Exercise Sheet 5: Matrices

Tutorial route (▸, in this order, about 80 minutes): 1.4, 2.2(a,b), 2.3(a,b), 2.4(a,b), 2.5(a,b), 2.7(a–c), 2.8(a–f). Everything else is for home study.

Levels: ⋆ drill, ⋆⋆ core, ⋆⋆⋆ challenge. Give exact answers unless a decimal is asked for.

1 Warm-up drills

Matrix algebra

Exercise 1.1 *(⋆ drill — addition, scalar multiple and product).* Let $\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 3 & -1 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} 0 & 4 \\ -2 & 5 \end{pmatrix}$.

Compute (a) $\mathbf{A}+\mathbf{B}$, (b) $2\mathbf{A}-\mathbf{B}$, (c) $\mathbf{AB}$.

Exercise 1.2 *(⋆ drill — which products are defined).*

(a) Matrices have sizes $\mathbf{A}: 2\times 3$, $\mathbf{B}: 3\times 3$, $\mathbf{C}: 3\times 1$, $\mathbf{D}: 1\times 2$. For each product $\mathbf{AB}$, $\mathbf{BA}$, $\mathbf{BC}$, $\mathbf{CD}$, $\mathbf{DA}$, $\mathbf{AC}$, state whether it is defined and, if so, its size.

(b) Compute $\mathbf{PQ}$ for $\mathbf{P} = \begin{pmatrix} 1 & 0 & 2 \\ -1 & 3 & 1 \\ 0 & 2 & 1 \end{pmatrix}$ and $\mathbf{Q} = \begin{pmatrix} 2 & 1 \\ 0 & -1 \\ 1 & 0 \end{pmatrix}$, stating its size first.

Exercise 1.3 *(⋆ drill — transpose and trace).* For $\mathbf{A} = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 0 \end{pmatrix}$ write down $\mathbf{A}^{\mathsf{T}}$ and compute $\operatorname{tr}(\mathbf{A})$.

Determinants and inverses

Exercise 1.4 *(▸ ⋆ drill — 2×2 determinants, then an inverse).* Let $\mathbf{M} = \begin{pmatrix} 3 & 5 \\ 1 & 2 \end{pmatrix}$ and $\mathbf{N} = \begin{pmatrix} 4 & 2 \\ 6 & 3 \end{pmatrix}$.

(a) Evaluate $\det \mathbf{M}$ and $\det \mathbf{N}$. Which matrix is invertible?

(b) Find the inverse of the invertible one and verify that its product with the original matrix is $\mathbf{I}$.

Exercise 1.5 *(⋆ drill — cofactor expansion).* Evaluate $\det \begin{pmatrix} 2 & 0 & 1 \\ 3 & 1 & -1 \\ 0 & 2 & 4 \end{pmatrix}$ by cofactor expansion along the first row.

Linear systems

Exercise 1.6 *(⋆ drill — solve by the inverse).* Solve by the inverse method: $3x+5y=1,\ x+2y=1$. (You may reuse $\mathbf{M}^{-1}$ from Exercise 1.4.)

Exercise 1.7 *(⋆ drill — solve by elimination).* Solve by elimination (write the augmented matrix and reduce): $x+y=5,\ 2x-y=1$.

Transformations

Exercise 1.8 *(⋆ drill — transformation matrices).* Write down the 2×2 matrices for (a) a rotation by $90°$ anticlockwise, (b) reflection in the x-axis, (c) an enlargement by factor 3. Then apply the rotation to $\mathbf{v} = (2, 1)$.

2 Tutorial problems

Exercise 2.1 *(⋆⋆ core — $\mathbf{AB} \neq \mathbf{BA}$).* Let $\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$.

(a) Compute $\mathbf{AB}$ and $\mathbf{BA}$ and confirm they differ.

(b) Verify $(\mathbf{AB})^{\mathsf{T}} = \mathbf{B}^{\mathsf{T}}\mathbf{A}^{\mathsf{T}}$ for these matrices.

Exercise 2.2 *(▶ ⋆⋆ core — data science: shop revenue).* A shop records units sold of three products on two days as the rows of $\mathbf{S} = \begin{pmatrix} 4 & 2 & 1 \\ 0 & 3 & 5 \end{pmatrix}$ (rows = days, columns = products). The unit prices are $\mathbf{p} = (3, 5, 2)^{\mathsf{T}}$.

(a) Explain why $\mathbf{Sp}$ is defined and what its entries mean.

(b) Compute the daily revenue vector $\mathbf{Sp}$.

Exercise 2.3 *(▶ ⋆⋆ core — the threaded system, part 1).* Let $\mathbf{A} = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 5 & 6 & 0 \end{pmatrix}$.

(a) Show $\det \mathbf{A} = 1$.

(b) Find $\mathbf{A}^{-1}$ by the cofactor (adjugate) method and verify $\mathbf{AA}^{-1} = \mathbf{I}$.

We reuse this matrix in Exercise 2.4.

Exercise 2.4 *(▶ ⋆⋆ core — the threaded system, part 2: two roads).* Solve $\mathbf{Ax} = \mathbf{b}$ with $\mathbf{A}$ from Exercise 2.3 and $\mathbf{b} = (6, 5, 11)^{\mathsf{T}}$, that is

$$\begin{aligned} x + 2y + 3z &= 6, \\ y + 4z &= 5, \\ 5x + 6y \quad &= 11, \end{aligned}$$

twice: (a) by the inverse method, and (b) by Gaussian elimination on the augmented matrix. Confirm the two answers agree.

Exercise 2.5 *(▶ ⋆⋆ core — infinitely many solutions, then none).*

(a) Solve, by elimination,

$$x + 2y - z = 3, \qquad 2x + y + z = 3, \qquad x - y + 2z = 0,$$

and write the solution set as the vector equation of a line (Unit 4). How does the echelon form reveal that the system has infinitely many solutions?

(b) Change the right-hand side of the third equation from 0 to 1 and apply the same row operations. What does the echelon form tell you now?

Exercise 2.6 *(⋆⋆ core — spot the case).* Apply elimination to

$$x + y + z = 1, \qquad x + 2y + 3z = 2, \qquad 2x + 3y + 4z = 5,$$

and explain what the echelon form tells you about the solution set.

Exercise 2.7 *(▶ ⋆⋆ core — physics: rotate then reflect).* A force $\mathbf{F} = (2, 3)$ (in newtons) is first rotated 90° anticlockwise and then reflected in the x-axis.

(a) Find the single 2×2 matrix $\mathbf{M}$ representing "rotate, then reflect".

(b) Compute the transformed force $\mathbf{MF}$.

(c) What single geometric transformation is $\mathbf{M}$?

Exercise 2.8 *(▶ ⋆⋆ core — conceptual, true or false).* Decide whether each statement is true or false; give a one-line reason or counterexample, and for (e) a proof.

(a) Matrix multiplication is commutative.

(b) Every square matrix has an inverse.

(c) $(\mathbf{AB})^{\mathsf{T}} = \mathbf{A}^{\mathsf{T}}\mathbf{B}^{\mathsf{T}}$.

(d) If $\det \mathbf{A} = 0$ then $\mathbf{Ax} = \mathbf{b}$ cannot have a unique solution.

(e) For all 2×2 matrices, $\operatorname{tr}(\mathbf{AB}) = \operatorname{tr}(\mathbf{BA})$. (Prove it with $\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, $\mathbf{B} = \begin{pmatrix} p & q \\ r & s \end{pmatrix}$.)

(f) There is a 2×2 matrix $\mathbf{A}$ with $\mathbf{Ax} = \mathbf{x} + \begin{pmatrix} 1 \\ 2 \end{pmatrix}$ for every $\mathbf{x} \in \mathbb{R}^2$.

3 Further practice (home)

Matrix algebra

Exercise 3.1 *(⋆⋆ core — data science: Markov chains (foreshadows Unit 6)).* A simple weather model has two states, *sunny* (S) and *rainy* (R). The *transition matrix* (columns = today, rows = tomorrow)

$$\mathbf{P} = \begin{pmatrix} 0.8 & 0.4 \\ 0.2 & 0.6 \end{pmatrix}$$

reads: if today is sunny, tomorrow is sunny with probability 0.8 and rainy with 0.2 (top and bottom of the first column), and similarly the second column handles a rainy today. Today is sunny, so the state vector is $\mathbf{x}_0 = (1, 0)^{\mathsf{T}}$.

(a) Compute $\mathbf{x}_1 = \mathbf{P}\mathbf{x}_0$ and say what its entries mean.

(b) Compute $\mathbf{P}^2$, then $\mathbf{x}_2 = \mathbf{P}^2\mathbf{x}_0$. Interpret the $(1, 1)$ entry of $\mathbf{P}^2$ (the probability of being sunny two days after a sunny day).

Determinants and inverses

Exercise 3.2 *(⋆⋆ core — when is it singular?).* For which value(s) of k is each matrix singular?

$$\text{(a) } \begin{pmatrix} 1 & k \\ k & 4 \end{pmatrix}, \qquad \text{(b) } \begin{pmatrix} 1 & 2 & 3 \\ 0 & k & 1 \\ 0 & 0 & 2 \end{pmatrix}.$$

Exercise 3.3 *(⋆⋆ core — inverse of a triangular matrix, then a system).* Let $\mathbf{U} = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$.

(a) Find $\mathbf{U}^{-1}$ by row-reducing $(\,\mathbf{U} \mid \mathbf{I}\,)$, and verify $\mathbf{U}\mathbf{U}^{-1} = \mathbf{I}$.

(b) Hence solve $x + y + z = 4,\ y + z = 2,\ z = 1$ by the inverse method.

Linear systems

Exercise 3.4 *(⋆⋆ core — Gaussian, then Gauss–Jordan).* Consider $x + y + z = 6,\ 2x - y + z = 3,\ x + 2y - z = 2$.

(a) Solve by Gaussian elimination (echelon form, then back-substitution).

(b) Continue the row operations from your echelon form to *reduced* row-echelon form (leading 1s, zeros above and below each pivot) and read the solution off directly.

Exercise 3.5 *(⋆⋆ core — a one-parameter solution set).* Find all solutions of $x+y+z = 3,\ x-y+z = 1$ and write the answer as the vector equation of a line. Then use your echelon form to decide whether the three columns of the coefficient matrix are linearly independent, and find a dependence between them.

Exercise 3.6 *(⋆⋆ core — physics: circuit currents).* Mesh analysis of a two-loop circuit gives the currents I_1, I_2 (in amperes) from

$$5I_1 - 2I_2 = 1, \qquad -2I_1 + 4I_2 = 6.$$

Solve by the inverse method.

Transformations

Exercise 3.7 *(⋆⋆ core — composing transformations, area and orientation).* Let $\mathbf{D} = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$ (a horizontal stretch) and $\mathbf{R} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ (rotation by 90°).

(a) Find the matrix for "stretch first, then rotate" and its determinant. Compute the other order $\mathbf{DR}$: does the order matter?

(b) On graph paper, sketch the unit square with corners $(0,0), (1,0), (1,1), (0,1)$ and its images under $\mathbf{D}$ and under $\mathbf{RD}$. Find the area of each image and check that it equals |det| times the original area.

(c) Repeat (b) for $\mathbf{F} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ and $\mathbf{N} = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$. For each, compute the determinant, state the effect on area and orientation, and say whether the map is invertible.

Challenge / extension

Exercise 3.8 *(⋆⋆⋆ challenge — inverse of a product).* Suppose $\mathbf{A}$ and $\mathbf{B}$ are invertible $n \times n$ matrices. Prove that $\mathbf{AB}$ is invertible with $(\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}$.

Exercise 3.9 *(⋆⋆⋆ challenge — rotations add (links to Units 1 and 3)).* Let $\mathbf{R}(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$.

(a) Explain geometrically why $\mathbf{R}(\theta)\mathbf{R}(\phi)$ must equal $\mathbf{R}(\theta + \phi)$.

(b) Compute $\mathbf{R}(\theta)\mathbf{R}(\phi)$ and, by comparing entries with $\mathbf{R}(\theta+\phi)$, recover the angle-addition formulae for cos and sin (stated in Unit 1). Compare with $e^{i\theta}e^{i\phi} = e^{i(\theta+\phi)}$ (Unit 3).

Exercise 3.10 *(⋆⋆⋆ challenge — matrices commuting with* $\mathbf{J}$*).* Find all 2×2 matrices $\mathbf{X}$ that commute with $\mathbf{J} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$, i.e. $\mathbf{XJ} = \mathbf{JX}$.